

# Varad Kulkarni

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## Education

### Vishwakarma Institute of Technology Pune

Bachelor of Technology in Computer Science (GPA: 8.61 / 10.00)

Aug 2022 - May 2026

Pune, Maharashtra

- **Relevant Coursework:** DSA (Java), MERN Stack, Computer Networks, DBMS, LLM's

## Experience

### Indian Institute of Tropical Meteorology (IITM)

Research Intern

Pune, Maharashtra

July 2025 - Present

- Working on Fine-Tuning large language Models (LLM's) to interpret and synthesize complex weather datasets for improved climate analysis and forecasting.

### BMC Software

Technical Analyst Intern

Pune, Maharashtra

January 2025 - June 2025

- Worked on Linux, Networking, Docker, and SQL to analyze technical issues and deliver solutions.

## Publications/Achievements

- Winner (3rd Place): Business and System Innovation Challenge 2024, Indonesia: **Developed an anti-plagiarism software prototype for Binus University, Indonesia**
- Patent Published for **Hybrid Routing Emergency Mesh Network System For Disaster Communication**: A decentralized communication system enabling real-time connectivity using hybrid mesh routing.
- Scopus Publication for the Paper Titled "Image Forgery Detection Using Machine Learning": **8th International Conference on Smart Trends in Computing and Communications (SmartCom'24) - 2024**

## Projects

### AcademAI | React.js, FastAPI, MongoDB, Python SDK, Gemini-2.0-Flash

- A full-stack research paper generator powered by LLMs that takes a GitHub repository link and user prompt as input, analyzes the codebase, and generates a structured research paper.
- Designed for students and researchers to automate technical documentation and research drafting.

### Mesh-Up | ESP8266, Java, Android Studio

- A decentralized mesh network enabling communication without traditional infrastructure like Wifi and Mobile Data, using IoT devices, laptops, and mobiles. It features UDP-based messaging, rebroadcasting, and flood protection for reliable and efficient communication.
- Designed for disaster recovery, remote areas, and events, it ensures real-time connectivity when infrastructure is unavailable.

## Technical Skills

**Languages:** React Native, C, Java, SQL, Python, Git, Arduino, Android, Solidity

**Web Development:** MERN Stack, JavaScript, HTML, CSS

**Operating Systems / Frameworks:** CentOS Linux, Docker (Containerization and Deployment)

## Volunteering Experience

### Microsoft Learn Students Club VIT Pune

Core Board Member and Web Developer

Pune, Maharashtra

January 2024 - May 2025

- Organized a 24-hour offline National hackathon at VIT Pune.
- Built an NGO website to enhance outreach and streamline donations.